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Investigating the factors affecting the adoption of cloud accounting in Saudi Arabia's small and medium-sized enterprises (SMEs)

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ABSTRACT

Small and medium-sized businesses (SMEs) can get new insights and comparative advantages from implementing information systems (ISs). This study aims to examine the factors that influence SMEs willingness to use cloud-based accounting. A theoretical framework was developed that incorporates technological and organizational variables from the Technology Organization Environment (TOE) framework as external factors for the Technology Acceptance Model (TAM). Additionally, there were proposed environmental factors that influence the adoption of cloud accounting. A survey was carried out among accountants employed in SMEs in Saudi Arabia. 354 valid responds were collected and analyzed. The data was analyzed with Structural Equation Modeling with the aid of the SmartPLS software. The results indicate that factors such as relative advantage, compatibility, complexity, organizational resources, employee capability, top senior management support, mimetic pressures, normative pressures, perceived usefulness, and perceived ease of use significantly impact the intention to adopt cloud accounting. The model acts as a blueprint for achieving favorable results in business organizations and offers suggestions for successful cloud accounting integration. This study utilizes two information technology (IT) adoption models to improve prediction accuracy, acting as a strategy for leadership to help businesses make well-informed decisions about cloud accounting despite resource constraints and apprehensions about cloud service utilization.

1. Introduction

Organizations can obtain enhanced competitiveness and financial advantages by effectively utilizing information and communications technologies (Nurcahyo and Putra, 2021; Phuthong, 2022). In the realm of cloud computing, IT innovation and investment play a crucial role (Lee, 2021), aiming to optimize the efficiency of IT resources. Cloud computing stands out as a significant technological advancement in recent years, representing the second-largest wave in Internet technology and poised to fuel future business growth. Cloud computing usage empowers organizations to effectively compete in the present economic environment. This system is a distributed architecture of interconnected computers provided by one or more suppliers through service level agreements. (Al-Okaily et al., 2022; Tawfik et al., 2022). Diverging from traditional computing requirements, cloud computing eliminates the need for users to store data on their devices or employ a plethora of complex programs (Ercan, 2010). Users gain advantages from having unlimited cognitive ability to process information, enabling them to access data at any given moment and from any place that is connected to the internet (Moudud-Ul-Huq et al., 2020; Yau-Yeung et al., 2020). The infrastructure of this model relies on three core technologies: web services, virtualization technology, and network-based computing (Armbrust et al., 2010). Cloud computing offers individuals and companies various advantages, including reduced automation and computing costs, decreased infrastructure maintenance expenses, and efficient management with user-friendly access (Ma et al., 2021).

At present, the global market for cloud computing is valued at around $350 billion, with projections anticipating its growth to reach approximately $650 billion by 2025. It is foreseen that 80 % of all organizations will have migrated to the cloud by that time (Flexera, 2020). Cloud computing has become a cornerstone of digital business, often referred to as the "silver bullet" by Okai et al. (2014). Despite its significance, many companies face challenges in optimizing this valuable tool. Cloud computing has revolutionized how businesses perceive,

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utilize, and pay for computing resources, ushering in a paradigm shift (Ma et al., 2021; Flexera, 2020; Yau-Yeung et al., 2020). This system allows third-party cloud service providers to deliver Internet-accessible IT resources through subscription or pay-per-use models, departing from traditional in-house IT deployment methods (Ma et al., 2021).

Cloud services' growing importance is evident in their remarkable expansion, offering an efficient means to leverage IT resources, generate employment, and contribute to environmental protection. The entrepreneurial landscape has evolved to become more inclusive during crises like the COVID-19 pandemic, introducing heightened uncertainty that entrepreneurs may find daunting (Alharbi, 2022). Although COVID-19 had a severe influence on the corporate realm, cloud computing emerged as a significant beneficiary. Due to the pandemic, the worldwide cloud market is expected to rise at an annual rate of 16.2 %, reaching $54.1 billion by 2027, up from $25.6 billion in 2022. (Open Source Services Market, 2022).

Each new addition and innovation in accounting has made it more capable of resolving challenges and providing users with satisfaction over the past few decades. Throughout its history, accounting has undergone gradual changes that led to its current state of technological progress. Technology has improved accounting to make it easier and more user-friendly by replacing manual methods with technological alternatives. Accounting has been able to serve users better since it has shifted towards faster technologies. In their paper, Mohanty and Mishra (2017) suggest that the cloud accounting model provides concurrent access to current financial statements online for business owners, accountants, auditors, and customers, etc. Technology can enable accountants to function more globally and drive companies to higher levels of proficiency if they allow it to prove its value. Furthermore, cloud computing enables enterprises to process data along the value chain activities (financing, sales, distribution and customer service) (Rababah and Al-nassar, 2020; Tawfik et al., 2022). Much research has discussed cloud computing as a new tool that can provide many strategic and operational advantages for those who adopt it. Interestingly, there is limited research on cloud accounting, most of which has attempted to explain the theoretical problems of cloud computing, the advantages and disadvantages, etc. (Adjei et al., 2021; Al-Okaily et al., 2022; Moudud-Ul-Huq et al., 2020; Özdemir and Elitaş, 2015; Senarathna et al., 2018; Tawfik et al., 2022). Moreover, the majority of studies on cloud accounting were conducted outside Saudi Arabia.

SMEs in Saudi Arabia have always been crucial in driving economic growth., constituting nearly 90 % of businesses, with approximately 85 % following a single proprietorship ownership structure (Saudi Gazette, 2017). As part of the Vision 2030 economic blueprint, the Saudi government recognizes the contribution of SMEs toward becoming one of the world's major economies by 2030, aiming for a 35 % GDP contribution from the sector. This growth spans various SME sectors, including manufacturing, food, telecommunications, and finetech. Despite being a developing country, Saudi Arabia faces challenges in its cloud services sector, aligning with its unique characteristics in terms of size, history, culture, information infrastructure, and economic structure (Alharbi, 2022). With Vision 2030 emphasizing digital transformation and innovation leadership (Adjei et al., 2021), There is an increasing need for SMEs in Saudi Arabia to embrace cloud computing technology, particularly in cloud accounting (Adjei et al., 2021). Reports and studies suggest that the utilization of cloud computing and accounting computing in the Arab world is not fully maximized. (Al-Okaily et al., 2022; Rababah and Al-nassar, 2020), motivating this study to address the knowledge gap, focusing specifically on Saudi Arabia

The literature suggests that challenges related to the adoption of cloud accounting can be categorized into three groups (Al-Sharafi et al., 2023; Bello et al., 2021). Firstly, there is a technological challenge associated with cloud computing, characterized by its intricate structure and compatibility concerns. Potential challenges may occur in terms of aligning with organizational technical aspects and the customization of existing applications for cloud systems (Tawfik et al., 2022). Implementing cloud solutions poses a significant challenge, requiring the seamless movement of applications or data without encountering compatibility issues (Gangwar et al., 2015). Integrating existing systems with cloud services during implementation can be complex, often requiring skills that may not always be readily available (Bello et al., 2021).

Secondly, cloud accounting presents organizational challenges, necessitating senior management to consider changes in workplace culture, processes, and relationships (Yau-Yeung et al., 2020). The essential factor in embracing cloud accounting is the top management's willingness to comprehend the business advantages and competitive edge it offers, which necessitates a significant financial commitment in technological proficiency (Lee, 2021). The implementation of a cloud accounting solution demands specialized human resources with the requisite skills and knowledge (e.g., employees with computer skills, IT specialists) (Tawfik et al., 2022). It is also vital that IT managers must provide the employees with cloud accounting training so that cloud accouning can be implemented effectively and improve employees' performance to achieve business goals. Having sufficient organizational resources enables them to accomplish their responsibilities easily or more efficiently.

Thirdly, cloud accounting introduces competitive challenges in the marketplace. Understanding the competitiveness of cloud computing for industry/market survival is a crucial aspect of adoption (Ma et al., 2021). Cloud service providers play a role in facilitating cloud accounting adoption, with challenges revolving around customer availability and support, as well as concerns about security in terms of third-party data access, transmission, and storage Existing adoption frameworks face challenges in the context of cloud accounting due to its intricate technology (Moudud-Ul-Huq et al., 2020). It is challenging to use any existing adoption framework in the cloud accounting context, because cloud accounting is an intricate technology (Tawfik et al., 2022).

It is argued that the TOE framework established by Tornatzky et al. (1990) and TAM established by Davis (1989), might best explain the adoption of cloud accounting in enterprises. Nevertheless, as discussed earlier, technological and organizational factors affect perceived ease of usefulness which means "the degree to which a person believes that using a particular system would be free of effort" (Davis, 1989), and perceived usefulness, which means "the degree to which a person believes that using a particular system would enhance his or her job performance" (Davis, 1989) of cloud accounting. This in turn affects the level of cloud accounting acceptance. Moreover, environmental factors such as availability, reliability, and security concerns influence cloud accounting adoption. There are three main constructs of the TOE framework while Perceived Usefulness (PU) and Perceived Ease of Use (PEU) are the core variables of the TAM model. It is important to consider the usefulness of cloud services because when users believe that their business can benefit from them in terms of efficiency, performance, and productivity, only then will they accept cloud accounting, and when they have accumulated enough knowledge about it and the resources required (Bello et al., 2021). Technology adoption studies have successfully deployed the TAM model for predicting and explaining users' intentions across a wide range of contexts. Thus in this research study, the basic model to be utilized will be TAM, as well as some constructs derived from the TOE framework that are incorporated into the proposed model. This will make it more relevant to cloud accounting as a whole.

Moreover, in the literature only a few studies have examined cloud accounting adoption using traditional adoption frameworks, including Tawfik et al. (2022), who used the TOE framework while Abu Afifa et al. (2022) chose the TAM model. Thus, in these studies the themes of cloud accounting adoption, including relative advantage, compatibility, competitive pressure, firm size, top management support, security, trust and privacy, infrastructure, external competing support perceived benefits, social influence, and business partner pressure were assessed

(Al-Okaily et al., 2022; Lin and Chen, 2012; Tawfik et al., 2022). Even though it has been proven that it delivers operational and strategic advantages, and enhances processes (Low et al., 2011; Tawfik et al., 2022), cloud accounting adoption is not growing as fast as anticipated (Özdemir and Elitaş, 2015; Soni et al., 2018; Tawfik et al., 2022), In previous research on the adoption of cloud accounting new technologies, security requirements, and future anticipations have been analyzed in relation to adoption of cloud accounting in these emerging milieu. These include technical issues which include resources that provide functionality, flexibility, and scalability, as well as human IT resources that result from training, experience, and insight (Moudud-Ul-Huq et al., 2020; Tawfik et al., 2022). It follows that specific aspects of particular organizations - relevant factors that affect the adoption of cloud accounting – must be addressed. Therefore, this study develops and examines a combined TAM-TOE model that seeks to better predict the factors that guide the usage of cloud accounting services in the organization.

To summarize, this paper aims to answer the following research question: What are the main determinants that impact the implementation of cloud accounting in SMEs in Saudi Arabia? In doing so, this paper revolves around three objectives:

1. Investigate the adaptive factors related to cloud accounting in SMEs.
2. Identify the connection between technological and organizational factors and their impact on perceived usefulness, influencing the intentions to adopt cloud accounting.
3. Examine and disclose the environmental influences affecting the adoption of cloud accounting.

By establishing these objectives, this paper attempts to contribute to the research on the determinants of Saudi Arabian SMEs' adoption of cloud accounting. Moreover, this work has explored and discussed the status of the entire development and examination of a combined TAM-TOE model that seeks to better predict the factors that guide the usage and especially the effectiveness of cloud accounting services. Prior tudies in cloud accounting were conducted employing qualitative methods (Ma et al., 2021; Yau-Yeung et al., 2020) and those researchers have called for more research to be done using a quantitative approach (Yau-Yeung et al., 2020) of existing practices. Finally, this work aims to call attention to some important challenges (Tawfik et al., 2022) and explore the factors influencing the intention of SMEs to embrace cloud accounting that can be used by government or industry policymakers planning cloud accounting strategies to assess the feasibility of this process

The remaining sections of this paper are as follows. Section 2 covers the literature review. Section 3 sets out theoretical framework and hypotheses development. Section 4 sets out the methodology for carrying out this analysis. Section 4 discusses the results, Section 5 explains the theoretical and practical implications, and Section 6 explains the limitations and future research directions. Finally the conclusion is articulated in Section 7.

2. Literature review

ISs research relies on a range of adoption and diffusion theories to understand, explain, and forecast the acceptance and integration of new technologies by individuals or organizations. Although these theories exhibit commonalities in their content and objectives, they diverge in terms of their actual implementations. (Al-Sharafi et al., 2023; Huang et al., 2022; Mujalli et al., 2022; Tawfik et al., 2022). They examine the adoption and diffusion of emerging technologies from various perspectives, aiming to elucidate the reasoning, scope, and processes involved in technology adoption and implementation (Chatterjee et al., 2021). Adoption theories analyze the elements that influence the acceptability of technology and examine these factors in detail (Henderson et al., 2012). Maruping et al. (2017) argue for giving more importance to behavioral expectations and actual usage as indicators of IT utilization in businesses, highlighting the need to move away from depending primarily on the behavioral intention paradigm.

Despite the growing interest in cloud accounting, there are several weaknesses and limitations in the existing literature. Dimitriu and Matei (2014) provide a comprehensive discussion on the risks and benefits of cloud adoption, but do not delve into specific accounting practices. Tarmidi et al. (2014) investigate the level of awareness and adoption of cloud computing by accountants in Malaysian SMEs, but only provide insights into one specific country and do not explore the reasons for the lack of knowledge. Similarly, Țugui and Gheorghe (2014) study the need for accounting model changes due to digitization, but their results are limited to the opinions of the survey respondents. Ebenezer et al. (2014) focus on the perceived benefits of cloud computing for accounting but do not provide empirical evidence to support their claims. Özdemir and Elitaş (2015) examine the infrastructure needs of cloud accounting systems but do not consider the implications of data privacy and security. Khanom (2017) provides an overview of cloud accounting but does not address the specific challenges and risks faced by accounting professionals. Additionally, the studies by Mohanty and Mishra (2017), Al-Zoubi (2017), Modi (2018), Rao et al. (2017), Soni et al. (2018), and Yau-Yeung et al. (2020) offer valuable insights into the benefits and risks of cloud accounting, but their findings are limited to specific contexts and do not provide a comprehensive understanding of the challenges and opportunities of cloud accounting systems. In summary, while the existing literature on cloud accounting provides valuable insights into its potential for accounting, it is limited by its concentration on particular contexts and absence of generalizable empirical evidence. Future research should aim to address these limitations by conducting more comprehensive research that considers the implications of cloud accounting for the challenges and opportunities faced by accounting professionals in different contexts. Thus, the current study investigates the factors influencing the intention of SMEs to adopt cloud-based accounting in Saudi Arabia.

3. Theoretical framework and hypotheses development

3.1. The technology-organizational-environmental (TOE) framework

Tornatzky et al. (1990) introduced the TOE framework as a model to elucidate various forms of IS/IT adoption at the organizational level. In the assessment of technology adoption, utilization, and value creation stemming from technological innovation, TOE holds a distinct advantage over alternative models due to its incorporation of technological, organizational, and environmental variables This framework has gained widespread acceptance within the academic community (I. Lee, 2021; Martins and Picoto, 2020; Tawfik et al., 2022)

Empirical applications and testing of this framework have provided enhanced insights into the adoption of technological advancements such as electronic data interchange and ISs. Technological factors encompass the attributes of both existing and emerging technologies, while organizational factors encompass scope, management support, culture, structural complexity, human capital quality, and available resources Environmental factors account for external influences, including competitors, business partners, cultural considerations, government incentives, and the presence of supportive technological infrastructure, all of which can either facilitate or impede business operations (Henderson et al., 2012; Martins and Picoto, 2020; Tawfik et al., 2022). This framework's applicability is not constrained by industry or firm size, offering a holistic view of technology adoption, implementation processes, anticipation of challenges, and their impact on value chains. It is crucial to consider the determinants guiding businesses' decisions when embracing new technologies (Chatterjee et al., 2021).

3.2. Technology acceptance model (TAM)

Hong & Tam (2006) state that the TAM is widely recognized as a very successful theoretical framework for understanding the adoption and usage of IT. The model facilitates the forecasting of an individual's adoption and utilization of IT in a professional environment (Low et al., 2011), by identifying the factors that influence acceptance (Zalazar et al., 2017). The TA) elucidates the connection between the intention to utilize a technology and its acceptability, emphasizing the importance of PU and PEU as key factors influencing the adoption of the system.

The TAM combines the Theory of Reasoned Action (TRA) and the Theory of Planned Behavior (TPB) to offer a clear and perceptive understanding of users' behavior. Davis's technology acceptance model posits that the acceptance of new technology depends on the PU and PEU (Davis, 1989). The TAM model posits that external variables exert an influence on users' views of the usefulness and ease of use of novel technologies. This influence, in turn, indirectly affects users' desire to adopt these technologies and shapes their attitudes towards them. Perceived usefulness refers to an individual's belief in a technology's capacity to improve their performance, while perceived ease of use relates to the factors that determine user acceptance, indicating the extent to which users perceive a new technology as easy to adopt without requiring significant physical effort or a difficult learning process. The TAM model considers the impact of external variables on consumers' perceptions to be negligible.

3.3. Developing a combined TAM-TOE framework

This study examines two well-known models for technology adoption: the TAM and the TOE framework. These models are routinely used to analyze how organizations adopt technology. Although the TAM model and TOE framework have proven its significance, supremacy, and pertinence in elucidating individual-level technology adoption through various empirical and conceptual investigations, they nevertheless possess certain constraints. The TAM model's two elements, PU and PEU, were found to explain around 40 % of system usage, according to Chatterjee et al. (2021). Nevertheless, the specific details regarding the external variables in the extended TAM model are still ambiguous. However, the TOE framework lacks certain primary components (Wang et al., 2010) and has been critiqued for its broadness (Al Nahian Riyadh et al., 2009). Researchers have recommended integrating the TOE framework with models that have well-defined constructs in order to boost the predictive accuracy of model outcomes. Nevertheless, the process of combining TOE and TAM is difficult because of the discrepancies in the factors and their importance.

Addressing these challenges, this study takes an approach that incorporates significant and insignificant variables from various studies on TOE and TAM to construct a new model. In contrast to the results of Jeyaraj et al. (2006), this study contends that, considering the contextual differences such as technology, study country, and company size (e.g., number of staff, turnover), no factor should be disregarded. Therefore, the objective of the study is to investigate the determinants that impact the inclination to embrace cloud accounting in SMEs in Saudi Arabia, employing the TAM as the foundational model. The exogenous factors influencing usage intention and efficiency are derived from prior studies on individual behaviors and are aligned with the TOE framework. PU and PEU are considered essential factors that are linked to the usage and behavioral intention. The pertinent adoption characteristics related to cloud accounting are integrated into the proposed conceptual framework, as seen in Fig. 1.

3.4. Relative advantage

Relative advantage, as described by Rogers (1983), refers to the degree to which a technology aspect is seen as offering superior advantages for enterprises. The adoption of a technology inside a workplace is heavily influenced by the advantages it offers in comparison to other choices. Cloud accounting offers significant comparative

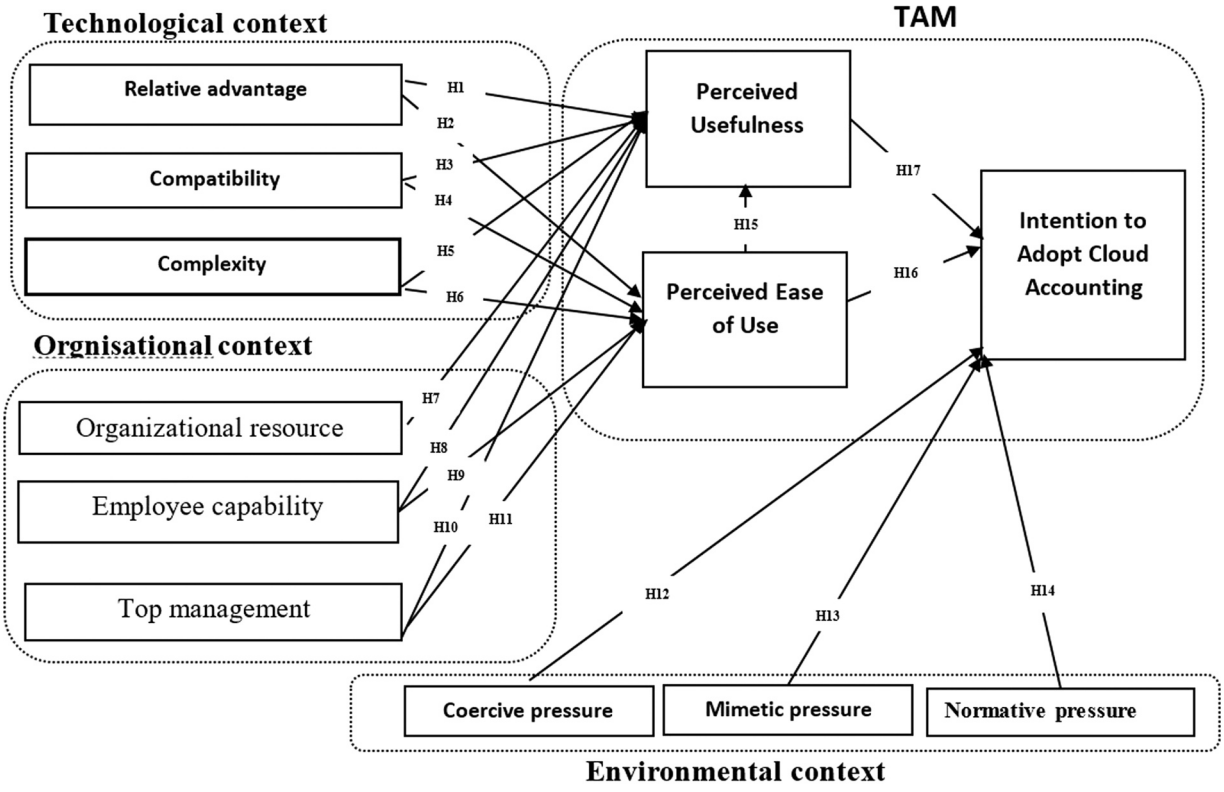


Fig. 1. Conceptual Model Proposal
Adapted from: Davis (1989); and Tornatzky et al. (1990).

advantages over traditional solutions due to its cost reduction, scalability, flexibility, mobility, and resource sharing capabilities. Scholars recognize the significance of relative advantage when assessing the effects of new technologies, particularly in the process of adoption (Gangwar et al., 2015; Tawfik et al., 2022). Therefore, when SMEs perceive innovation as having a greater advantage, they are more likely to adopt it. This component has exhibited a robust correlation with both relative advantage and technological adoption in prior research (Azmi et al., 2016; Henderson et al., 2012). Given this reasoning, two hypotheses are put forth:

H1. The relative advantage has a beneficial impact on the PU in relation to the intention to embrace cloud accounting.

H2. The relative advantage has a beneficial impact on the PEU when it comes to the intention to use cloud accounting.

3.5. Compatibility

The ability to change procedures after adopting a new IS can lead to improved information quality, fewer errors, and increased efficiency (Azmi et al., 2016; Chatterjee et al., 2021; Gangwar et al., 2015). Perceived compatibility, defined by Chatterjee et al. (2021), is the extent to which an innovation aligns with potential users' existing values, previous experiences, and requirements. Compatibility considers an organization's ability to reconcile new technologies and innovations with existing values, behavioral patterns, and experiences (Bhattacherjee & Lin, 2015). Previous research emphasizes compatibility as a crucial factor in adopting novel IS innovations (Low et al., 2011; Tawfik et al., 2022). Numerous studies on IT adoption highlight the validity of compatibility in *PEU* and PU (Chatterjee et al., 2021; Katebi et al., 2022). Building on the positive association between compatibility and new technology adoption found in previous studies (Chatterjee et al., 2021; Gangwar et al., 2015), two hypotheses are posited:

The ability to change procedures after adopting a new IS can lead to improved information quality, reduced errors, and higher efficiency (Azmi et al., 2016; Chatterjee et al., 2021; Gangwar et al., 2015). Perceived compatibility, as described by Chatterjee et al. (2021), refers to the degree to which an invention corresponds with the values, past experiences, and needs of potential customers. Compatibility refers to an organization's capacity to harmonize new technologies and innovations with preexisting values, behavioral patterns, and experiences (Bhattacherjee & Lin, 2015). Prior studies highlight the significance of compatibility in the adoption of new IS improvements (Low et al., 2011; Tawfik et al., 2022). Multiple research on the adoption of IT emphasize the importance of compatibility in determining the *PEU* and PU (Chatterjee et al., 2021; Katebi et al., 2022). Based on the favorable correlation discovered in earlier research (Chatterjee et al., 2021; Gangwar et al., 2015) between compatibility and the adoption of new technology, two theories are proposed:

H3. Compatibility exerts a positive impact on the PU in relation to the inclination to embrace cloud accounting.

H4. Compatibility exerts a positive impact on the PEU in relation to the inclination to use cloud accounting.

3.6. Complexity

The level of complexity of an innovation is ascertained by the degree of difficulty involved in its execution (Rogers, 2003). Companies that encounter challenges in utilizing IT may face difficulties in effectively improving or creating new operational procedures. Complexity in cloud accounting adoption is assessed based on factors such as the duration of task execution, integration of applications with specialized cloud infrastructure, efficiency of data transfer, and system functionality. Despite being sometimes seen as separate entities, intricacy and ease of use are intricately interconnected. Prior research has shown that there is a favorable connection between complexity and the adoption of new technology (Azmi et al., 2016; Henderson et al., 2012). This implies that there is an opposite relationship between complexity and both the perceived ease of use and PU of the technology (Chatterjee et al., 2021; Gangwar et al., 2015). Given this line of thinking, two theories are put forward:

H5. Complexity has a detrimental impact on users PU to adopt cloud accounting.

H6. Complexity has a detrimental impact on the PEU when it comes to the intention to use cloud accounting.

3.7. Organizational resources

Organizational resources, as defined by managers' perceptions and evaluations, encompass how aware, committed, and governed an organization is in embracing IT (Katebi et al., 2022; Tawfik et al., 2022). Technology competence refers to the proficiency in utilizing an organization's internal technology assets, such as IT infrastructure and internet capabilities (Martins and Picoto, 2020). The adoption of electronic data interchanges by small enterprises is heavily influenced by the perceived availability of technology resources. The comprehensive definition encompasses two aspects: financial preparedness (availability of funds for implementing and sustaining cloud computing) and technological preparedness (availability of skilled personnel and infrastructure for managing and utilizing cloud computing). Research has demonstrated a favorable correlation between the availability of resources inside an organization and the adoption of new technology (Chatterjee et al., 2021; Martins and Picoto, 2020). The effectiveness of technologies can be enhanced by businesses with robust infrastructure, knowledgeable employees, and financial support. Thus, the following hypothesis is proposed:

H7. Organizational resources exert an advantageous effect on the PU in relation to the inclination to embrace cloud accounting.

3.8. Employee capability

Employees have a crucial impact in shaping the results of their workplaces, acting as invaluable assets for companies (Chatterjee et al., 2021). The effective adoption of new technology hinges on the presence of knowledgeable and competent personnel (Soni et al., 2018). Successfully integrating a new technology requires the organization's human resources to possess the expertize necessary for its effective implementation (Bello et al., 2021). Given the complexity of cloud accounting as an IS, organizations need to invest in training and educating their staff. The purpose of this training is to improve employees' comprehension and approval of the advantages of cloud computing, while also addressing any concerns they may have about its utilization. Prior research has repeatedly emphasized a favorable correlation between staff competence and the use of novel technologies. (Chatterjee et al., 2021; Martins and Picoto, 2020). Moreover, such training initiatives not only mitigate ambiguity but also empower employees to develop the knowledge required for effective usage in the future. Keeping these considerations in mind, two additional hypotheses are proposed:

H8. Employee capability has a beneficial impact on the PU of cloud accounting adoption intention.

H9. Employee capability has a beneficial impact on the PEU related to the intention to use cloud accounting.

3.9. Top management support

The unequivocal endorsement of new technology adoption by a company's top or senior management is paramount for establishing a competitive and market-dominant environment (Adjei et al., 2021; I. Lee, 2021). This endorsement enables businesses to surmount internal challenges and resistance to change. Emphasizing the role of an owner/manager as a positive agent for change in the workplace and support for technology acceptance (Tawfik et al., 2022). According to Salwani

et al. (2009), the views and behaviors of senior executives on the effectiveness of technical advancements play a crucial role in generating value for the firm. This entails fostering a long-term vision, reinforcing values, allocating resources, effective resource management, cultivating a favorable workplace climate, enhancing individual self-efficacy, and providing support to overcome obstacles and resistance to change, as observed by Ramdani et al. (2009). Existing literature consistently suggests a positive correlation between the support of senior management and the PU and ease of use PEU when adopting new technologies. Therefore, the subsequent hypotheses are suggested for empirical examination:

H10. The support of top management has a positive impact on the PU in relation to the intention to embrace cloud accounting.

H11. The support of top management has a positive impact on the perception of how easy it is to utilize cloud accounting, which in turn affects the intention to adopt it.

3.10. Coercive pressure

Business-to-business coercion entails utilizing formal or informal methods to compel other companies to conform to specific behaviors (Rogers, 2003). Consequently, powerful institutions leverage their authority or influence to induce entities with lesser authority to voice concerns about specific policies and/or actions, aiming to enhance credibility and gain an advantage (Alshirah et al., 2021; Cleary & Quinn, 2016). Coercive pressure can originate from multiple sources, such as parent firms, policy requirements, and resource-abundant businesses, which are integrated within exchange relationships (Teo et al., 2003). Numerous studies have illustrated that coercive organizations or corporations dominating a market or industry can exert pressure on other companies, compelling them to comply with demands for overall dominance. The exertion of coercive coercion has the capacity to impact the adoption of technology within businesses (Adjei et al., 2021; Alshirah et al., 2021). Given this assumption, the subsequent hypothesis is proposed:

H12. Coercive pressure has a beneficial impact on the inclination to embrace cloud accounting.

3.11. Mimetic pressure

An organization might emulate another successful company within its industry, replicating its practices. As outlined by Teo et al. (2003), mimetic pressure manifests in two ways. Firstly, if numerous organizations adopt a cloud accounting system, it becomes more prevalent in the industry, leading to diminished innovation and reduced competition. Secondly, organizations may imitate the practices of successful competitors adopting cloud accounting systems to maintain competitiveness. Previous studies have demonstrated that mimetic pressure can impact technology adoption (Adjei et al., 2021; Alshirah et al., 2021; Azmi et al., 2016). Considering this, the following hypothesis is presented:

H13. Mimetic pressure positively influences the inclination to use cloud accounting.

3.12. Normative pressure

According to DiMaggio and Powell (1983), affiliates, businesses, professional associations, or networks exert normative pressure based on shared values. A profession is collectively viewed by its members, setting out the appropriate codes of conduct. Professional networks and associations influence organizations to adopt technologies, perpetuating normative principles about certain behaviors (Azmi et al., 2016; Henderson et al., 2012). Normative pressures demonstrate how professional norms and cultures affect an organization, shaping the requirements that companies must adhere to in order to comply with professional standards and valid procedures. Normative pressures do not influence businesses through coercion; instead, they drive compliance, given that decision-makers play a crucial role in industrial and technical firms. Given that normative pressure can affect technology adoption in organizations (Adjei et al., 2021; Alshirah et al., 2021), therefore proposing the following hypothesis:

H14. Normative pressure postively infleunce the inclination to use cloud accounting.

3.13. Perceived usefulness

The perception of ease of use refers to an individual's belief regarding the level of effortlessness associated with using a new system or technology (Davis, 1989). TAM has been effectively utilized in numerous studies to clarify an individual's willingness to adopt new technologies or systems (Lee et al., 2015). This study utilizes the TAM to elucidate the goals of using cloud accounting in companies. It focuses on the probability that an application system will improve the performance of a potential user in an organization. Thus, the following hypotheses are proposed:

H15. PU has postively influences on the inclination to embrace cloud accounting.

3.14. Perceived ease of use

Consistent with the TAM, PEU serves as a predictor of users' intentions to utilize novel systems or technologies. According to Yousafzai et al. (2007), if a system is user-friendly, it is more likely to be embraced by people. The TAM model is widely recognized as the most acknowledged framework for predicting individuals' intentions to adopt new technologies (Venkatesh and Davis, 2000). Studies have delved into artificial intelligence, such as Chatterjee et al. (2021), while multipurpose information appliances were explored by Hong and Tam (2006) to forecast the inclination to embrace a new system. From these discussions, it becomes evident that the intention to adopt cloud accounting hinges on its ease of use. The TAM model suggests that PU is influenced by PEU, as easy-to-use technologies hold higher value (Al-araibi et al., 2019; Chatterjee et al., 2021; Gangwar et al., 2015). Consequently, the following hypotheses are proposed:

H16. PEU positevely influences the intention to adopt cloud accounting.

H17. PEU positevely effects PU.

4. Methodology

4.1. Measurement instrument

This study utilized a quantitative methodology to investigate the factors that impact the adoption of cloud accounting in SMEs in Saudi Arabia. The hypotheses and conceptual model were validated using a quantitative research method, which required the creation of a questionnaire. The identification of constructs was derived from a comprehensive examination of existing literature and the application of the theoretical framework. All items representing these constructs were incorporated into the questionnaire for the proposed research modelIn the pretesting phase, questions representing each construct were assessed for meaning and context by 15 experts consisting of practitioners and academic researchers who possess expertize in cloud-based services, namely accounting services. Some items were modified to improve the simplicity, adequacy, and clarity of the study's measurements.

The questionnaire was translated from English to Arabic, as Arabic is the native language of Saudi Arabia. Two Saudi linguist teachers and two academics from Saudi universities validated the translation process. To ensure clarity and ease of understanding, the research adopted a 'back translation' method, a widely used practice in international research, to validate the quality of translation and address any relevant issues (Ozolins, 2008; Saunders et al., 2019). This method validates the

quality of translation and confirms that relevant issues have been accounted for (Ozolins, 2008).

The final questionnaire comprised 45 items covering all pertinent constructs, and respondents provided their answers using a 5-point Likert scale, with '1' representing severe disagreement and '5' indicating strong agreement. Table 1 provides a comprehensive description of the measurement device.

4.2. Participants and data collection process

The research study employed a purposive sample method, targeting on employees within accounting departments with existing knowledge or interest in cloud accounting. Qualified participants were individuals working in SMEs in Saudi Arabia who demonstrated a good understanding of cloud accounting. It was a voluntary study, and respondents were asked to send the online link through a variety of communication channels from late July to early December 2022, such as mailing lists, personal e-mails, social media groups, etc. A cover letter accompanied the survey, outlining the study's purpose, emphasizing the concept of voluntary involvement, and informing participants about the research objectives and their right to withdraw at any stage, either before or after completing the questionnaire. Because the target population's size is unclear, determining the precise number of individuals meeting the sampling criteria for the study was challenging.

The sample size holds significance in ensuring a representative and appropriately sized sample for conducting accurate statistical analyses. Previous studies have proposed different sample sizes, and various theories exist to determine an adequate sample size (Oliveira et al., 2011). However, sample size requirements differ based on the statistical analysis type and the literature presents a range of perspectives, even when employing similar techniques (Hair et al., 2013). In order to achieve a sufficient sample size (with a margin of error of ±5 %, a proportion of 0.05, and a confidence level of 0.95) and considering the lack of exact knowledge about the number of units representing the study population or its potentially large size, the determined acceptable sample size was 322. Consequently, 354 completed surveys were deemed valid for analysis. The online survey questionnaire comprised three sections. The first section included an illustration of cloud accounting, and subsequent questions were accessible only to those who viewed the illustration and met the appropriate qualification. The subsequent sections captured demographic information, including gender, age, educational background, employment experience, annual sales, company age, and the number of employees. Likert five-point scales were employed to measure the variables incorporated into the framework.

4.3. Data analysis

The data analysis comprised two stages: firstly, descriptive statistics were applied to outline the demographic features of the respondents; secondly, inferential statistics were employed to analyze the responses. Hair et al. (2013) Advocated for use the PLS-SEM method, especially beneficial for small sample sizes, as it enables testing a very intricate model with mediation analysis. In PLS-SEM studies, Henseler and Chin (2010) advocated a two-stage analysis of the measurement and structural models. Assessment of the measurement model involved conducting reliability tests, examining convergent validity, and evaluating discriminant validity (Hair et al., 2013; Ringle et al., 2015). The evaluation of structural models involved significant T-values, coefficients of determination (R2), and f2 effect sizes. The Q2 index was used to test predictive power, while the model's overall fit was evaluated using Square Root Mean Residual (SRMR) and Goodness-of-Fit (GOF). As mentioned earlier, the software utilized for this analysis is SmartPLS (Ringle et al., 2015).

Table 1
Measurement items.

Constructs	Code	Sample of measurement item	Source
Relative advantage	RA1	"When I use cloud accounting, I am able to scale up my work when and as required".	(Chatterjee et al., 2021; Nurcahyo and Putra, 2021; Rogers, 2003)
	RA2	"When I use cloud accounting, I can access information from any time and from anywhere".	
	RA3	"When I use cloud accounting, it reduces the operational costs".	
	RA4	"When I use cloud accounting, we do not need to maintain the IT infrastructure".	
Compatibility	CO1	"In case of any incompatibility issue, we ask the cloud service provider to offer integrated services".	(Chatterjee et al., 2021; Géczy et al., 2012; Henderson et al., 2012; Rogers, 2003)
	CO2	"Cloud accounting is compatible with the existing technological architecture of my company".	
	CO3	"Customization in cloud accounting is easy".	
	CO4	"Cloud accounting is compatible with the existing IT infrastructure".	
Complexity	CX1	"Cloud accounting information system is flexible enough to interact with".	(Armbrust et al., 2010; Chatterjee et al., 2021; Gangwar et al., 2015)
	CX2	"Using cloud accounting informs me about the vulnerability of computer breakdowns and loss of data".	
	CX3	"When we use cloud accounting, we find it difficult to integrate my existing work with the cloud-based services".	
	CX4	"When we perform a lot of tasks together, using cloud accounting takes up too much of my time".	
Organizational resource	OR1	"Our firm has financial resources to support cloud accounting usage".	(Armbrust et al., 2010; Chatterjee et al., 2021)
	OR2	"Our firm has IT resources to support cloud accounting usage".	
	OR3	"Our firm is willing to provide training on cloud accounting for employees".	
	OR4	"Our firm is ready to provide technical expertize to support cloud accounting adoption".	
Employee capability	EC1	"Our employees are IT literate".	(Chatterjee et al., 2021)
	EC2	"Our employees' understanding of cloud accounting is very good".	
	EC3	"Our firm has at least one employee who is a cloud accounting expert".	

(*continued on next page*)

Table 1 (*continued*)

Constructs	Code	Sample of measurement item	Source
	EC4	"Our employees have enough knowledge to use the results produced by cloud accounting".	
Top management Support	TMS1	"Top management is aware of the benefits of cloud accounting".	(Chatterjee et al., 2021; Tawfik et al., 2022)
	TMS2	"Top management is highly interested in cloud accounting".	
	TMS3	"Top management gives strong support for cloud accounting usage".	
	TMS4	"Top management provides adequate financial resources for implementing cloud accounting".	
Coercive pressure	CP1	"Our trading partners are pressuring our firm to adopt cloud accounting".	(Azmi et al., 2016; Henderson et al., 2012; H.-H. Teo et al., 2003)
	CP2	"Our suppliers are pressuring our firm to adopt cloud accounting".	
	CP3	"Financial institutions are pressuring our firm to adopt cloud accounting".	
Mimetic pressure	MP1	"Our firm mimics competitors and adopts cloud accounting".	(Henderson et al., 2012; Mohammad Fadzillah et al., 2020)
	MP2	"Our firm's pressure leads to competitors mimicking us and adopting cloud accounting".	
	MP3	"Our firm's cloud accounting adoption has been influenced by what leaders in the industry have done".	
Normative pressure	NP1	"Our firm partners pressured our firm to use cloud accounting".	(Azmi et al., 2016; Henderson et al., 2012)
	NP2	"Regulators pressured our organization to adopt cloud accounting".	
	NP3	"Our external auditors will pressure our firm to use cloud accounting".	
Perceived Usefulness	PU1	"I agree that using cloud accounting makes our firm more efficient".	(Davis, 1989; Venkatesh and Davis, 2000)
	PU2	"I believe that the use of cloud accounting increases productivity in our organization".	
	PU3	"I can achieve things more quickly using cloud accounting".	
	PU4	"Cloud accounting reduce the work costs".	
Perceived Ease of Use	PEU1	"The process of using cloud accounting is easily understood by me".	(Daniel, 2011; Venkatesh and Davis, 2000; Zalazar et al., 2017)
	PEU2	"I will be able to use the cloud accounting in our organization".	
	PEU3	"I agree that all employees can quickly learn about the usage of cloud accounting".	
	PEU4	"It is easy to make use of cloud accounting".	

Table 1 (*continued*)

Constructs	Code	Sample of measurement item	Source
Intention to Adopt Cloud Accounting	IACA1	"Overall I think that using cloud accounting services is advantageous".	(Chatterjee et al., 2021; Davis, 1989; Gangwar et al., 2015)
	IACA2	"Overall, I am in favor of using cloud accounting services".	
	IACA3	"I would like to use cloud accounting to its full potential".	
	IACA4	"Overall, I think using cloud accounting will enhance our organization's productivity".	

4.4. Descriptive statistics

Table 2 presents the demographic characteristics of the respondents. 82.2 % of the participants are male, 17.8 % are female, and 42.7 % were aged between 30 and 39 years. Moreover, 51.1 % had worked for between 5 and 9 years while 74.9 % had a Bachelor degree. Most participants are from micro businesses and they make up 54.8 % of the study sample while 43.3 % work in small business. Lastly, 2 % work in medium-sized businesses. The bulk of participants are working in enterprises less than 5 years old and they constitute 53.1 % of the sample. Table 2 shows that all sample organizations utilize cloud accounting solely for internal purposes. A few programs are deployed, including Gmail, Google Drive, and the cloud marketing service, which rose dramatically after the COVID-19 pandemic. Saudi Arabian accountants are familiar with cloud computing (such that accounting precepts and its practices are consistent and will continue to be consistent). Instead of a manual or computerized system, cloud accounting for them is practical and user-friendly, data is accessible universally and at all times, it is more cost-effective and efficient, flexible, mobile, data storage is unlimited, etc.

5. Evaluation of the measurement model

The reflecting measurement model was used to calculate all

Table 2
Respondents' demographic information.

Characteristic	Description	Percentage	Frequency
Gender	Male	82.2 %	291
	Female	17.8 %	63
Age	Less than 29 years	31.6 %	112
	30 – 39 years	42.7 %	151
	40–49 years	21.5 %	76
	50 and more	4.2 %	15
Education	Diploma or below	22 %	78
	Bachelor degree	74.9 %	265
	Masters degree	3.1 %	11
	PhD	0 %	0
Employment	Less than 5 years	33.6 %	119
	5–9 years	51.1 %	181
	10–14 years	14.1 %	50
	15 or above	1.1 %	4
Annual sales (in Saudi riyals)	Less than 3 million	54.8 %	194
	3–40 million	43.2 %	153
	More than 40 million	2.0 %	7
Company age (years)	Less than 5 years	53.1 %	188
	From 5–10 years	40.1 %	142
	From 11–15 years	6.8 %	24
Number of employees	1–5 employees	50.3 %	178
	6–49 employees	43.2 %	153
	50–249 employees	6.5 %	23

constructions. Nomological validity was confirmed by comparing the correlation patterns of constructs with those anticipated by theory. (Hair et al., 2013; Henseler et al., 2015). The constructs, derived from the literature review and hypotheses, are considered valid from a nomological standpoint. Indicators were loaded onto the measurement model, and their significance was assessed (Hair et al., 2013). Following Hair et al.'s (2013) recommendation, indicators were required to have an outer loading above 0.7. Consistency and reliability of constructs were enhanced by eliminating indicators with a loading between 0.4 and 0.7 (Henseler et al., 2015; Henseler & Chin, 2010). Table 3 groups constructs based on the indicators they represent. The reliability of the reflected measurement model was evaluated by calculating Cronbach's alpha, Composite Reliability (CR), Rho-a, and evaluating construct commonality. Convergent validity was assessed by the Average Variance Extracted (AVE), whereas divergent validity was examined through the heterotrait-monotrait test (HTMT) and the Fornell-Larcker criterion. The Q2 index was utilized to measure the predictive validity of the measurement model.

5.1. Convergent validity

Factor loadings are deemed satisfactory when they exceed 0.5 (Hair et al., 2013), as illustrated in Table 3. Notably, the factor loadings for most constructs surpass 0.7, with a few exceeding 0.6, affirming the reliability of the study's measures. According to Hair et al. (2013), the reliability of Cronbach's alpha estimates is assessed by examining intercorrelations. The criteria outlined by Hair et al. (2013) stipulate that Cronbach's alpha should exceed 0.7. The values presented in Table 3 confirm this criterion, demonstrating that the measurement model exhibits consistent internal consistency. Alternative metrics like CR, which assess the outer loadings of indicators, can be used due to the constraints of Cronbach's alpha. (Hair et al., 2013; Mujalli et al., 2022; Ringle et al., 2015). A CR value greater than 0.7 is recommended (Hair et al., 2013), and Table 3 substantiates the reliability of the measurement model based on this criterion. Dijkstra and Henseler (2015) introduced the Rho-a index, akin to CR but considering the weight of each construct rather than the outer loading of its respective indicator. An acceptable threshold for this coefficient is above 0.7 (Hair et al., 2013), and all constructs in Table 3 meet this criterion.

The AVE is calculated by averaging the squared loadings of each indicator. (Hair et al., 2013). A suitable AVE value should not fall below 0.5 (Hair et al., 2013), and Table 3 demonstrates that the AVE for all constructs exceeds this threshold, validating the measurement model's dependability based on this index. Another test for convergent validity involves comparing CR with AVE, where CR values should surpass AVE for each construct (Hair et al., 2013; Henseler & Chin, 2010; Kline,

Table 3
Reliability, validity and quality of the measurement model.

Constructs	Items	Loadings	Cronbach's Alpha	rho_A	Composite Reliability	AVE
Relative advantage	RA1	0.84	0.78	0.84	0.85	0.59
	RA2	0.75				
	RA3	0.84				
	RA4	0.64				
Compatibility	CO1	0.79	0.81	0.83	0.87	0.63
	CO2	0.85				
	CO3	0.78				
	CO4	0.75				
Complexity	CX1	0.70	0.81	0.85	0.87	0.63
	CX2	0.88				
	CX3	0.78				
	CX4	0.81				
Organizational resources	OR1	0.85	0.86	0.87	0.91	0.71
	OR2	0.84				
	OR3	0.84				
	OR4	0.83				
Employee capability	EC1	0.71	0.78	0.81	0.86	0.60
	EC2	0.72				
	EC3	0.82				
	EC4	0.84				
Top management support	TMS1	0.86	0.83	0.84	0.89	0.66
	TMS2	0.80				
	TMS3	0.76				
	TMS4	0.82				
Coercive pressure	CP1	0.85	0.82	0.83	0.89	0.73
	CP2	0.85				
	CP3	0.86				
Mimetic pressure	MP1	0.87	0.72	0.77	0.84	0.64
	MP2	0.84				
	MP3	0.68				
Normative pressure	NP1	0.90	0.78	0.81	0.87	0.70
	NP2	0.83				
	NP3	0.77				
Perceived ease of use	PEU1	0.83	0.82	0.86	0.88	0.66
	PEU2	0.86				
	PEU3	0.87				
	PEU4	0.66				
Perceived usefulness	PU1	0.79	0.77	0.76	0.85	0.59
	PU2	0.80				
	PU3	0.82				
	PU4	0.65				
Intention to adopt cloud accounting	IACA1	0.82	0.83	0.86	0.89	0.67
	IACA2	0.85				
	IACA3	0.89				
	IACA4	0.69				

2011). This condition is satisfied for all constructs (refer to Table 3). Additionally, communality, equivalent to the square root of each construct's outer loadings in a standardized indicator, measures the extent to which the construct explains variation in an item (Hair et al., 2013; Ringle et al., 2015). Commensurate with acceptable standards, a communality value should not dip below 0.5 (Hair et al., 2013), and as evident in Table 3, each construct exhibits an AVE above 0.5, thus confirming communality.

5.2. Discriminant validity

It has also been reported that the Fornell-Larker criterion is effective for assessing convergent validity (Fornell and Larcker, 1981). A matrix is provided in this test using the primary diameter of the square root of the AVE values for each build (Fornell and Larcker, 1981; Hair et al., 2013). For a construct, the square root of the AVE should be higher than its correlation with another construct in this assessment. The Fornell-Larker matrix confirms convergent validity as depicted in Table 4.

As well as cross-loading test problems, Henseler et al. (2015) proposed HTMT to solve Fornell-Larcker criterion problems. This index should have a minimum threshold of 0.9 for all correlations, according to Henseler et al. (2015). Looking at Table 5, different constructs have their own values for this index. Based on these results, convergent validity is confirmed since correlations among different constructs amount to less than 0.9.

5.3. Structural model tests

A significance test was employed to assess the reliability and validity of the structural model, after confirming the validity and reliability of the measurement model, it is important to consider both the robustness and orientation of hypotheses. Subsequently, model fit indices and the quality of the structural model were assessed, focusing solely on direct effects in our model. We used bias-corrected 95 % confidence intervals to analyze the hypotheses concerning direct impacts.

Hypothesis H1 was supported, with RA demonstrating a beneficial and significant effect of on PU (β =.155, t = 2.921, p =.004). Consistent with this, H3 was supported, as CO exhibited a positive and significant impact on PU (β =.192, t = 2.997, p =.003). Likewise, H5 gained support, with Complexity (CX) having a positive and significant impact on PU (β =.244, t = 3.604, p <.001). H7 was strongly supported, indicating that OR had a positive and significant impact on PU (β =.300, t = 4.197, p <.001). EC was found to have a beneficial and significant influence on PU (β =.192, t = 2.624, p =.009), supporting H8. TMS also demonstrated a positive and significant effect on PU (β =.248, t = 4.600, p <.001), confirming H10. H15 were validated, as PEU exerted a positive and significant impact on PU (β =.240, t = 4.511, p <.001).

Furthermore validating H2, RAhad a positive and significant impact on Perceived Ease of Use (PEU) (β =.154, t = 2.601, p =.010). CO positively and significantly impacted PEU (β =.219, t = 2.986, p =.003), supporting H4. However, CX did not have a significant impact on PEU (β =.013, t =.161, p =.872), leading to the non-support of H6. EC had a positive and significant impact on PU (β =.203, t = 2.975, p =.003), confirming H9. TM positively and significantly impacted PEU (β =.187, t = 3.050, p =.002), supporting H11. Hypotheses related to environmental factors were also examined. CP had no meaningful effect on the Intention to Adopt Cloud Accounting. (β =.026, t =.524, p =.601), leading to invalidate H12. MP had a notable and beneficial effect on the intention to embrace cloud accounting. (β =.140, t = 2.674, p =.008), confirming H13. Similarly, H14 gained support, as NP had a positive and significant impact on intention to adopt cloud accounting (β =.158, t = 3.176, p =.002). PEU had a favorable and substantial effect on the intention to use Cloud Accounting (β =.303, t = 5.663, p <.001), thus confirming H16. Moreover, PU positively and significantly influenced the inclination to adopt cloud accounting (β =.276, t = 4.911, p <.001), leading to the support of H17.

5.4. The coefficient of determination

The model's prediction ability was assessed by R^2, indicating the amount of variance explained by the predictors in the outcome variables. Cohen (1988) ranked the R^2 values as small (R^2 =.2), medium (R^2 =.5) and large (R^2 =.8). RA, CO, CX, OR, EC, TMS and PEU together explained a variance of 46.2 % (R^2 =.462) in PU. RA, CO, CX, EC and TMS together explained a variance of 13.5 % (R^2 =.135) in PEU. CP, MP, NP, PEU and PU together explained a variance of 30.5 % (R^2 =.305) considering the adoption of cloud accounting.

5.5. Predictive relevance of the structural model

The structural model's predictive accuracy was evaluated using the Q2 index, as suggested by Geisser. (1974). This index assesses the structural model's quality using the Blindfolding approach in its calculation (Ringle et al., 2015). According to Byrne (2013), a Q2 value of 0.02, 0.15, and 0.35 for an endogenous construct indicates poor, average, and strong predictive quality, respectively. The Cross Validated Redundancy(CV-RED) approach specifically assesses endogenous constructs, while Cross validated communality (CV-COM) examines all constructs. In Table 6, each Q2 indicator was found to be greater than zero, indicating high-quality structural modeling based on the data. Therefore, the predictive accuracy of the structural model was determined by the positive values of Q2 (> 0), signifying acceptable/good quality or predictive relevance. Chin's (1998) criteria for Q2, denoting 0.02, 0.15, and 0.35 as poor, average, and strong, respectively, are illustrated in Table 6.

6. Discussion

Cloud computing is the latest technology that businesses are now widely embracing. Currently, it is increasingly accepted in accounting for a number of reasons such as relative advantage, compatibility, complexity, having enough resources, employees' capability, top

Table 4
Discriminant validity: the Fornell-Larcker criterion.

	CO	CP	CX	EC	IACA	MP	NP	OR	PEU	PU	RA	TMS
CO	***0.794***											
CP	0.530	***0.854***										
CX	-0.646	-0.586	***0.794***									
EC	-0.399	-0.393	0.482	***0.774***								
IACA	0.123	0.174	-0.065	0.066	***0.817***							
MP	0.463	0.438	-0.381	-0.354	0.187	***0.801***						
NP	0.420	0.434	-0.484	-0.364	0.255	0.317	***0.834***					
OR	-0.440	-0.380	0.378	0.605	-0.013	-0.458	-0.461	***0.841***				
PEU	0.225	0.133	-0.105	0.059	0.452	0.094	0.237	-0.014	***0.810***			
PU	-0.116	-0.152	0.301	0.428	0.311	-0.213	-0.188	0.494	0.314	***0.766***		
RA	0.444	0.356	-0.440	-0.272	0.149	0.291	0.399	-0.120	0.219	0.083	***0.771***	
TMS	0.110	0.082	-0.017	-0.114	0.212	0.071	0.186	0.020	0.208	0.269	0.117	***0.812***

Table 5
Discriminant validity – HTMT ratios.

	CO	CP	CX	EC	IACA	MP	NP	OR	PEU	PU	RA	TMS
CO	-											
CP	0.648	-										
CX	0.781	0.722	-									
EC	0.499	0.512	0.598	-								
IACA	0.160	0.198	0.111	0.117	-							
MP	0.584	0.585	0.481	0.475	0.226	-						
NP	0.519	0.550	0.599	0.489	0.293	0.392	-					
OR	0.516	0.456	0.446	0.749	0.085	0.571	0.556	-				
PEU	0.274	0.152	0.185	0.127	0.549	0.144	0.279	0.083	-			
PU	0.303	0.212	0.309	0.477	0.405	0.244	0.246	0.544	0.432	-		
RA	0.612	0.499	0.594	0.411	0.171	0.408	0.508	0.257	0.253	0.266	-	
TMS	0.140	0.105	0.051	0.146	0.254	0.094	0.231	0.097	0.236	0.409	0.143	-

Table 6
Hypotheses Testing of Structural Model.

Construct	Estimate	T	P	R^2	Q^2	Status
RA → PU	0.155	2.921	0.004	0.462	0.245	H1: Supported
CO → PU	0.192	2.997	0.003			H3: Supported
CX → PU	0.244	3.604	0.000			H5: Supported
OR → PU	0.300	4.197	0.000			H7: Supported
EC → PU	0.192	2.624	0.009			H8: Supported
TMS → PU	0.248	4.600	0.000			H10: Supported
PEU → PU	0.240	4.511	0.000			H15: Supported
RA → PEU	0.154	2.601	0.010	0.135	0.077	H2: Supported
CO → PEU	0.219	2.986	0.003			H4: Supported
CX → PEU	0.013	0.161	0.870			H6: Not Supported
EC → PEU	0.203	2.975	0.003			H9: Supported
TMS → PEU	0.187	3.050	0.002			H11: Supported
CP → IACA	0.026	0.524	0.601	0.305	0.192	H12: Not Supported
MP → IACA	0.140	2.674	0.008			H13: Supported
NP → IACA	0.158	3.176	0.002			H14: Supported
PEU → IACA	0.303	5.663	0.000			H16: Supported
PU → IACA	0.276	4.911	0.000			H17: Supported

management support, PEU and PU, and the pressures that are described as coercive, mimetic and normative. All of these factors directly influence the decision to implement cloud accounting. The relative advantage is the extent to which firms view advanced technology as better than a given predecessor. It is considered to be an issue likely to affect the intention to embrace cloud accounting. It is evaluated based on both its perceived utility and simplicity of usage. The study's findings show that the relative advantage has a substantial impact on the perceived usefulness (H1) and perceived ease of usefulness (H2) in adopting cloud accounting. The outcome aligns with findings from other studies such as Abdekhoda et al. (2019) and Gangwar et al. (2015), which means the user believes how useful the technology will be (Davis, 1989). Morover, this study agrees with Musawa and Wahab (2012) who stated that the perceived advantages of the technology had the greatest influence on the decision to use it. Amoako-Gyampah and Salam (2004) found a positive correlation between the ERP system's relative advantage and its adoption, which aligns with the results of this study. Users should understand how cloud computing will increase their productivity and perceive the system as easy to use and relevant to their daily tasks.

Organizational compatibility evaluates how well innovation matches a company's current ideals, historical experiences, and future requirements. The findings indicate that compatibility significantly shapes perceived usefulness (H3) and perceived ease of usefulness (H4) in the adoption of cloud accounting. These results align with previous research by Chatterjee et al. (2021), Katebi et al. (2022), Lin and Chen (2012), and Wang et al. (2010). The positive correlation observed between compatibility and perceived ease of usefulness underscores that a high level of compatibility with cloud accounting enables users to seamlessly incorporate it into their job roles without necessitating significant adjustments to their tasks and work styles.

Compatibility significantly impacts perceived utility by aligning cloud capabilities with the current technical architecture, including format, interface, and other structural aspects. This highlights a robust relationship between compatibility and perceived usefulness in the context of cloud accounting usage. Particularly challenging for those managing complex applications and internal systems, ensuring compatibility with current infrastructure, policies, IT environments, and business processes and requirements becomes a pivotal task. To address this challenge, managers are advised to adapt existing processes to guarantee that cloud-based solutions harmonize with the organization's overarching framework (Lin and Chen, 2012).

The study's complexity is assessed based on data transmission efficiency, interface design, and system functionality. Interestingly, the findings show positive and negative outcomes of the complexity-related factors. Regarding the positive effects and this refers to complexity (H5), it exerts a significant influence on perceived usefulness to adopt cloud accounting. This finding is consistent with prior research (Chatterjee et al., 2021; Wang et al., 2010), and it means that cloud accounting is easier for employees as it takes less time to complete tasks. The finding of (H6) is that complexity negatively influences perceived ease of usefulness to adopt cloud accounting and it is in line with what other research (Azmi et al., 2016; Martins and Picoto, 2020; Ramdani et al., 2009) reported. Cloud accounting has a detrimental impact on corporate ideas regarding its ability to enhance work performance and user-friendliness. The less complexity there is in cloud accounting, the more probable staff members will be able to utilize it. So accounting systems developers should design it so that it provides users with a more intuitive experience and be more relevant to the specific job or task so that ease of use is promotedResearchers found that cloud accounting is not complicated if: firstly, One has the requisite skills to incorporate current applications with the cloud infrastructure.; secondly, has an understanding of the new technology; and thirdly, one is not afraid of changes to established business processes. By providing easy-to-use development tools, integrating data and systems and assuring the maintenance of each layer and their connections, service providers ease the anxiety associated with cloud computing in an organization (Lin and Chen, 2012).

Among the tested organizational factors, the finding for (H7) regarding &H7organization resources have a considerable impact on the perceived utility of cloud accounting., agrees with Gangwar et al., (2015), Musawa and Wahab (2012) and Wang et al. (2010). Organizations' ability to adopt cloud computing depends on their infrastructure and level of knowledge. Organizations with high preparedness levels are more inclined to employ cloud computing services. It is therefore crucial that managers and policymakers focus on physical infrastructure, intangible knowledge, and hiring IT-skilled staff as well as enhancing

intangible knowledge/resources. This needs to be in tandem with developing strategic projects to support business growth.

The finding for employee capability significantly affects the perceived utility (H8) and perceived ease of usefulness (H9) in relation to the adoption of cloud accounting.Those findings echo what other studies documented (Gangwar et al., 2015; Jain and Bhardwaj, 2010). Through training, employees gain first-hand knowledge and experience of cloud accounting from both a functional and technical perspective. They will become well-educated, knowledgeable, responsible, and experienced. In this way, they can readily utilize it and comprehend its significance for their task demands. Consequently, managers should create training courses to facilitate the efficient integration of cloud computing.

The finding for top management shows that it does have a significant influence on the perception of usefulness (H10) and perceived ease of use (H11). Those findings are in line with Low et al., (2011), Ramdani et al. (2009) and Teo et al. (2009). Thus, top management is crucial in influencing and inspiring individuals to perform their tasks more effectively. The company's executives demonstrate a commitment to continuously support an environment conducive to cloud computing (by supplying essential resources including time, space, equipment, and personnel). It is therefore usual practice for top management to adopt technology using a top-down approach, since it is essential for them to comprehend utilizing technology to enhance business performance, overcome perceived gaps, and capitalize on business opportunities.

According to the finding for (H12), it is revealed that coercive pressure is insignificant as far as the adoption of cloud accounting is concerned. The reason for this could be that the owners of businesses believe that this will put financial pressure on them. This finding is consistent with Amores-Salvadó et al. (2014). Previous research has demonstrated that coercive pressure can potentially undermine a business's competitiveness (Eiadat et al., 2008). Its impact on Saudi Arabia's SMEs' adoption proclivity is not yet really known. Only if policymakers try to enforce proper and flexible regulations on firms will coercive pressure be a successful driver of technology adoption in the long-term, according to several studies. Butler (2011) discovered that when policymakers impose stricter and enforceable regulations, the adoption of technologies increases.

A significant role was also played by mimetic pressure (H13) when adopting cloud accounting. This finding is in line with previous studies in technologies such as Leventis (2013) and Azmi et al. (2016). Briefly, imitation occurs when firms copy their competitors that are structurally similar (Azmi et al., 2016; DiMaggio and Powell, 1983). Organizations rely on the experience and success of competitors to avoid uncertainty and risk, leading to their intention to adopt cloud accounting to 'stay in the game'. Thus, businesses are more likely to implement cloud accounting when they perceive some advantages and in this way are able to compete. It is cheaper for decision-makers to copy IT choices used in another firms rather than create their own.

A significant influence from normative pressure (H14) was also found for environmental factors. The findings of this study are in line with earlier research that demonstrates how normative pressure is critical to the adoption of IT such as Alshirah et al. (2021) and Teo et al. (2003). A number of suppliers, government agencies, and other stakeholders are using cloud accounting to exert normative pressure. As a result of these organizations sharing norms, values, and information with firms, adopting cloud accounting is better integrated into business decision-making in terms of the environment. Organizations will need to adhere to professional standards and implement procedures and systems endorsed by various agencies due to normative pressures. The normative pressures stemming from the widespread use of that technology by several organizations affect SMEs to adopt IT/IS not because of their assessment of its potential efficiency but because of the pressures caused by the normative pressures themselves.

7. Research implications

7.1. Theoretical contributions

A theoretical assessment has been developed to explain the digitalization of sustainable manufacturing and production systems based on environmental, social, and technological contexts. It was hypothesized here that such digital systems are compatible with the trend to accept cloud accounting. Therefore, this study has applied the concept of cloud accounting in order to interpret the socio-environmental aspects as well as the technological factors involved. This has been achieved by devising a hybrid model - the integrated TOE-TAM model. In this hybrid model, organizations are explored from the perspective of social, environmental, and technological considerations in regard to their intention to adopt cloud accounting. To date, only a few studies have been conducted on how cloud accounting generates a competitive advantage (Ma et al., 2021). Here, it has been explained how the TOE framework will help to interpret how cultural factors block the adoption of a new technology (e.g., big data) in an entity (Lunde et al., 2019). Based on the findings documented in this study, it may have been helpful to use an updated standard adoption model. This study makes use of a TOE-TAM-based integrated model in order to select better antecedents. The proposed research model has been enriched by reviewing what other studies suggested. The purpose of presenting this theoretical model is to improve the digitalization of accounting systems in organizations by embracing cloud accounting (Davis, 1989). PEOU and PU served as mediating variables to trigger cloud accounting adoption intentions, so this study bridges a variety of factors covering social, environmental, and technological aspects with the intention of adopting AI technology.

7.2. Practical implication

The findings of this study could be important to IS users in Saudi Arabia because identifying technological, organizational, and environmental factors that influence the success of utilizing cloud accounting in a comprehensive framework is extremely beneficial. Furthermore, putting a spotlight on these factors makes it possible to better comprehend the conditions required for successful cloud accounting implementation. Based on the findings of this study, SMEs considering cloud accounting can determine whether their procedures, policies, staff, etc., are appropriate. Also, they can determine whether their technological innovation, organizational and IS-related factors are suitable. Managers are now in a better position to understand, assess, and explore various IT sources, and these can improve business performance. Currently, the overwhelming majority of Saudi managers lack adequate knowledge of cloud computing and need to devote more time to learn how it will benefit them (Assante et al., 2016).

The different viewpoints on cloud accounting discussed in this research constitute an administration strategy that assists enterprises in making goods decisions in implementing cloud accounting. This strategy also assists managers to determine the best aspects of cloud accounting services to invest in (Lee, 2021; Schneider and Sunyaev, 2016). Therefore, a detailed and multifaceted visualization of cloud acconting adoption can assist administrators to become acquainted with the options available (Al-Okaily et al., 2022; Senarathna et al., 2018; Tawfik et al., 2022). Strategies enabling cloud accounting execution necessitate a comprehensive plan for evaluating the actions to undertake.

8. Conclusion, limitations and avenues for future research

ISs have advanced greatly with the advent of cloud accounting. A number of appealing features of this model include its flexibility, scalability, and cost-effectiveness. To understand whether different industries adopt cloud accounting for the same reasons, the factors that explain cloud accounting adoption must be clarified and a careful analysis undertaken. As well as discussing potential benefits, the study

discusses an important aspect of cloud computing, which is cloud accounting, and explains how regulatory, environmental, and technological factors play a significant role in Saudi Arabian enterprises' acceptance of cloud accounting. Implementation of cloud accounting has not reached high levels despite the already noted technology's ability to create operational and strategic improvements and competitive advantage.

The purpose of this study was to investigate the factors influencing the adoption of cloud accounting in enterprises. Due to its extensive recognition in the literature on technology adoption, TAM was selected to establish the basis for assessing these factors in the existing literature. A TAM-TOE framework for cloud accounting adoption was developed due to the increased relevance of the TOE framework in recent literature. The TAM model was expanded by include a set of TOE variables related to cloud accounting adoption as external factors.The findings indicated that the compatibility, complexity, organizational resources, employee capability, top management support, mimetic pressures, normative pressures, perceived utility, and perceived ease of use significantly influence the inclination to use cloud accounting. Moreover, the TAM model and TOE framework were integrated using TOE variables as external variables. An empirical test was performed on the hypotheses, leading to the development of an original model by integrating two prominent adoption frameworks in a novel way. This study makes a unique contribution to the literature on cloud accounting and its findings should be validated in other contexts in future research. To date many companies in Saudi Arabia have adopted cloud accounting but more case studies are required to understand current and future trends for several industries. As companies develop more internet-based infrastructure, they can gradually increase the number and speed of operations using cloud accounting.

There are several limitations in this study. First, the search tool consisted of a questionnaire designed for SMEs' accountant executives who participated as key respondents in the survey. Although the research tool has already undergone validation and reliability testing, their opinions could still be biased in this regard. When external opinions are incorporated into the analysis, such as those of partners, customers, competitors, and suppliers, bias can be mitigated. The information provided by respondents needs to be validated through the evaluation of companies' annual reports. This study presents cross-sectional data since all the main elements were only collected at one time during a specific period. So in effect the data presented here is cross-sectional. It is crucial to consider the long-term impacts on the development of cloud accounting services, as well as the backing from senior management and competitive influences. As well, the data were collected from SMEs in only one country, Saudi Arabia. Cultural variations between developed and developing economies may impact the management of ISs and the perception of technology. Research frameworks need to be further researched and attached to samples from different countries in order to generalize ideas and review concepts. Cultural differences can influence individuals' perspectives on tasks associated with adopting new technology like cloud accounting. A more conclusive hypothesis could also be tested through further research. Future research could further investigate the connections between the highlighted elements in this study and other pertinent domains to overcome these constraints.

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Ethical Statement

This material is the authors' own original work, which has not been previously published elsewhere.

The paper is not currently being considered for publication elsewhere.

The paper reflects the authors' own research and analysis in a truthful and complete manner.

The paper properly credits the meaningful contributions of co-authors and co-researchers.

The results are appropriately placed in the context of prior and existing research.

All sources used are properly disclosed (correct citation). Literally copying of text must be indicated as such by using quotation marks and giving proper reference.

All authors have been personally and actively involved in substantial work leading to the paper, and will take public responsibility for its content.

I agree with the above statements and declare that this submission follows the ethical policies and requirements.

CRediT authorship contribution statement

Mohammad Jibran Gul Wani: Investigation, Methodology, Software, Writing – review & editing. **Ahmed Almgrashi:** Conceptualization, Data curation, Formal analysis, Writing – original draft. **Abdulwahab Mujalli:** Conceptualization, Formal analysis, Writing – original draft. **Tahany Khormi:** Validation, Visualization, Writing – review & editing. **Muath Qahtani:** Resources, Validation, Visualization.

Declaration of Competing Interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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